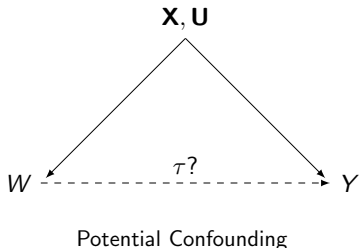


PLSC 504 – Fall 2026

Causal Inference with Observational Data

September 21, 2026

Causal Inference, Observational Data



Here:

- Y is the outcome of interest,
- W is the primary predictor / covariate ("treatment") of interest,
- T_i is the "treatment indicator" for observation i ,
- We're interested in estimating τ , the "treatment effect" of W on Y ,
- $\mathbf{X}$ are observed confounders,
- $\mathbf{U}$ are unobserved confounders.

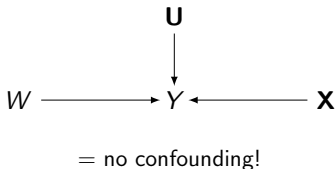
- **Randomize**

(or...)

- Selection on Observables:
 - Stratification
 - Regression / Weighting
 - Matching (including propensity scores)
- Instrumental Variables Approaches
- Regression Discontinuity Designs (“RDD”)
- Differences-In-Differences (“DiD”)*
- Synthetic Controls*
- Others...

* We'll discuss these approaches in a couple weeks, as models for panel/time-series cross-sectional data.

Under Randomization



Note:

- Randomized assignment of W “balances” covariate values – both observed and unobserved – *on average*...
- That is, under randomization of W :

$$E(\mathbf{X}_i, \mathbf{U}_i \mid W_i = 0) = E(\mathbf{X}_i, \mathbf{U}_i \mid W_i = 1)$$

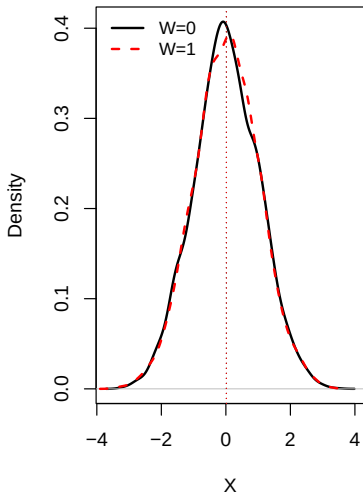
or, more demandinglly,

$$E[f(\mathbf{X}, \mathbf{U}) \mid W_i = 0] = E[f(\mathbf{X}, \mathbf{U}) \mid W_i = 1]$$

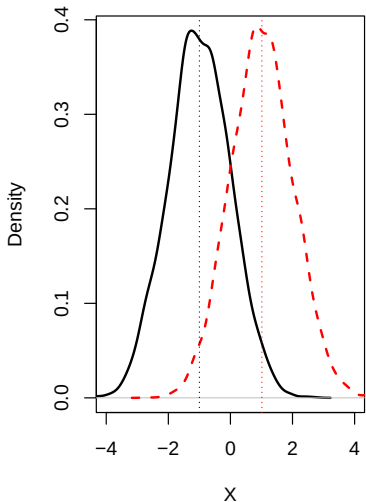
- Can yield imbalance by random chance...

Covariate Balance / Imbalance

Balanced X



Unbalanced X



Covariate Imbalance Under Randomization

Why seek balance when randomizing?

- More accurate estimates of treatment effects
- Higher statistical power

Possible Approaches:

1. Force balance by design:

- Stratification / blocking
- Matching / paired randomization (see below)
- Rerandomization approaches (e.g., [Morgan and Rubin 2012](#))

2. Post-randomization analysis:

- Pre- vs. post-treatment Y values / “gain scores”
- (Post-treatment) stratification by $\mathbf{X}$
- (Pre-treatment) covariate adjustment via weighting / regression

Nonrandom Assignment of W_i

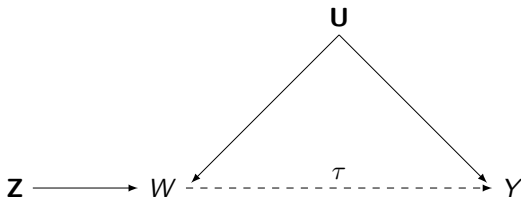
Valid causal inference requires $Y_{0i}, Y_{1i} \perp W_i | \mathbf{X}_i, \mathbf{U}_i$

- That is, treatment assignment W_i is *conditionally ignorable*

“What if I have unmeasured confounders?”

- In general, that's a bad thing.
- One approach: obtain *bounds* on possible values of τ
 - Assume you have one or more unmeasured confounders
 - Undertake one of the methods described below to get $\hat{\tau}$
 - Calculate the range of values for $\hat{\tau}$ that could occur, depending on the degree and direction of confounding bias
 - Or ask: How strong would the effect of the $\mathbf{U}$ s have to be to make $\hat{\tau} \rightarrow 0$?
- Some useful cites:
 - Rosenbaum and Rubin (1983)
 - Rosenbaum (2002)
 - DiPrete and Gangl (2004)
 - Liu et al. (2013)
 - Ding and VanderWeele (2016)

Digression: Instrumental Variables



Instrumental Variables

As in the more general regression case where we have $\text{Cov}(\mathbf{X}, u) \neq 0$, instrumental variables can be used to address confounding in causal analyses.

Instrumental Variables: Considerations

IV requires:

1. $\text{Cov}(\mathbf{Z}, W) \neq 0$
2. $\mathbf{Z}$ has no independent effect on Y , except through W
(“exclusion restriction”)
3. $\mathbf{Z}$ is exogenous [i.e., $\text{Cov}(\mathbf{Z}, \mathbf{U}) = 0$]

Arguably most useful when treatment compliance is uncertain /
driven by unmeasured factors (“intent to treat” analyses)

Turns out it's not that useful...

- [Bound et al. \(1995\)](#): Weak instruments are worse than endogeneity bias
- [Young \(2022\)](#): Inferences in published IV work (in economics) are wrong and terrible
- [Lal et al. \(2023\)](#): Same in political science
- [Mellon \(2024\)](#): Weather as an instrument? LOL
- [Shalizi \(2020, chapters 20-21\)](#): Gathers all the issues together, sometimes hilariously

Other (less doomerish) references:

- [Imbens et al. \(1996\)](#) (the overly-cited one)
- [Hernan and Robins \(2006\)](#) (making sense of things)
- [Lousdal \(2018\)](#) (a good intuitive introduction)

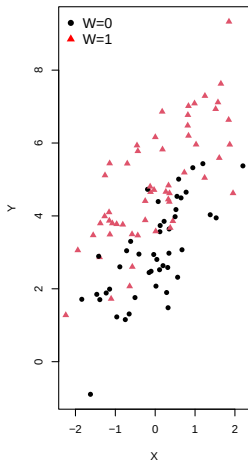
Nonrandom Assignment of W_i (continued)

So...

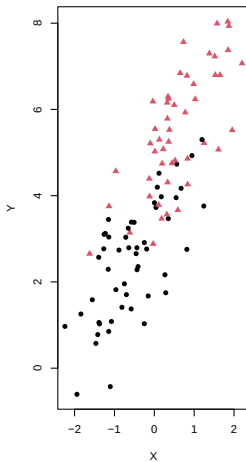
- Causal inference with observational data typically requires that $\mathbf{U} = \emptyset$...
- This typically requires a strong theoretical motivation in order to assume that the observed $\mathbf{X}$ exhausts the list of possible confounders.
- **Even if** this assumption is reasonable, there are two (related) important concerns:
 - Lack of *covariate balance* (as above)
 - Lack of *overlap* among observations with $W_i = 0$ vs. $W_i = 1$
 - The latter is related to *positivity*, the requirement that each observation's probability of receiving (or not receiving) the treatment is greater than zero

Overlap

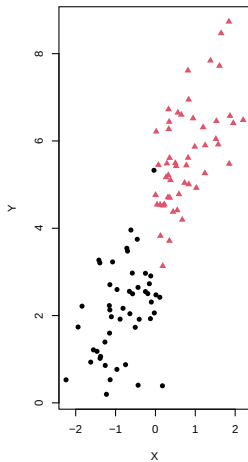
Complete Overlap



Moderate Overlap



No Overlap



In general:

- Ensuring overlap allows us to make counterfactual statements from observational data
 - Requires that we have comparable $W_i = 0$ and $W_i = 1$ units
 - It's *necessary* – no overlap means any counterfactual statements are based on assumption
 - Think of this as an aspect of *model identification* (Crump et al. 2009)
 - Most often handled via matching
- Ensuring covariate balance corrects potential bias in $\hat{\tau}$ due to (observed) confounding
 - This can be done a number of different ways: stratification, weighting, regression...
 - Key: Adjusting for (observable) differences across groups defined by values of W
- In general, we usually address overlap first, then balance...

Matching is a way of dealing with one or both of covariate overlap and (im)balance.

The process, generally:

1. Choose the $\mathbf{X}$ on which the observations will be matched, and the matching procedure;
2. Match the observations with $W_i = 0$ and $W_i = 1$;
3. Check for balance in $\mathbf{X}_i$; and
4. Estimate $\hat{\tau}$ using the matched pairs.

Variants / considerations:

- 1:1 vs. 1:k matching
- “Greedy” vs. “Optimal” matching (see [Gu and Rosenbaum 1993](#))
- Distances, calipers, and “common support”
- Post-matching: Balance checking...

- Simplest: Exact Matching

- For each of the n observations i with $W = 1$, find a corresponding observation j with $W = 0$ that has identical values of $\mathbf{X}$
- Calculate $\hat{\tau} = \frac{1}{n} \sum (Y_i - Y_j)$
- Generally not practical, especially for high-dimensional $\mathbf{X}$
- Variants: “coarsened” exact matching (e.g., [Iacus et al. 2011](#))

- Multivariate Matching

- Match each observation i which has $W = 1$ with a corresponding observation j with $W = 0$, and whose values on $\mathbf{X}_j$ are the most similar to $\mathbf{X}_i$
- One example: Mahalanobis distance matching, based on the distance:

$$d_M(\mathbf{X}_i, \mathbf{X}_j) = \sqrt{(\mathbf{X}_i - \mathbf{X}_j)' \mathbf{S}^{-1} (\mathbf{X}_i - \mathbf{X}_j)}.$$

Flavors of Matching (continued)

- Propensity Score Matching
 - Match observation i which has $W = 1$ with observation j having $W = 0$ based on the closeness of their *propensity score*
 - The propensity score is, $\Pr(W_i = 1|\mathbf{X}_i)$, typically calculated as the predicted value of T_i (the treatment indicator) from a logistic (or other) regression of T on $\mathbf{X}$.
 - The assumptions about matching [that Y is orthogonal to $W|\mathbf{X}$ and that $\Pr(W_i = 1|\mathbf{X}_i) \in (0, 1)$] mean that $Y \perp W | \Pr(T|\mathbf{X})$.
 - In practice: [read this...](#)
- Other variants: Genetic matching ([Diamond and Sekhon 2013](#)), “generalized full matching” ([Sävje et al. 2021](#)), etc.¹

¹[Shalizi \(2016\)](#) notes that “(A)pproximate matching is implicitly doing nonparametric regression by a nearest-neighbor method,” and that “(M)aybe it is easier to get doctors and economists to swallow “matching” than “nonparametric nearest neighbor regression”; this is not much of a reason to present the subject as though nonparametric smoothing did not exist, or had nothing to teach us about causal inference.”

Interestingly, quite a few of the good matching programs written for R have been written by political scientists...

- the `Match` package (does propensity score, M -distance, and genetic matching, plus balance checking and other diagnostics)
- the `MatchIt` package (for pre-analysis matching; also has nice options for checking balance)
- the `optmatch` package (suite for 1:1 and 1: k matching via propensity scores, M -distance, and optimum balancing)
- `quickmatch` (“quick” generalized full matching)
- `matching` (in the `arm` package)

Regression Discontinuity Designs

“RDD”:

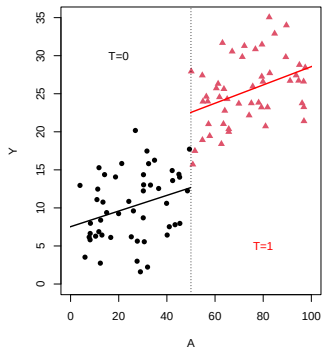
- Treatment changes abruptly [usually at some threshold(s)] according to the value(s) of some measured, continuous, pre-treatment variable(s)
 - This is known as the “assignment” or “forcing variable(s),” sometimes denoted **A**
 - Formally:

$$T_i = \begin{cases} 0 & \text{if } A_i \leq c \\ 1 & \text{if } A_i > c \end{cases}$$

- Intuition: Observations near but on either side of the threshold(s) are highly comparable, and can be used to (locally) identify τ
- This is because variation in T_i near the threshold is effectively random (a “local randomized experiment”)
- E.g. [Carpenter and Dobkin \(2011\)](#) (on the relationship between the legal drinking age and public health outcomes like accidental deaths)

RDD (continued)

- Pluses:
 - Can be estimated straightforwardly, as:
$$Y_i = \beta_0 + \beta_1 A_i + \tau T_i + \gamma A_i T_i + \epsilon_i$$
 - Generally requires fewer assumptions than IV or DiD (and those assumptions are easier to observe and test)
- Minuses:
 - Provides only an estimate of a local treatment effect
 - Fails if (say) subjects can manipulate A in the vicinity of c
- [Lee and Lemieux \(2010\)](#) is an excellent (if fanboi-ish) review
- R packages: `rddtools`, `rdd`, `rdrobust`, `rdpower`, `rdmulti`



Options:

- R
 - Packages for matching are listed above (Matching, MatchIt, quickmatch, etc.)
 - Similarly for RDD (rddtools, rdd, rdrobust, etc.)
 - IV regression: ivreg (in AER), tsls (in sem), others
 - See generally the [Causal Inference](#) and [Econometrics](#) CRAN Task Views
- There's a pretty good NumPy/SciPy-dependent package for Python, called (creatively) [CausalInference](#)
- Also: Stata has a large suite of routines for attempting causal inference with observational data

Example: Sports and Grades in High School

Question: Does participation in high school varsity sports help or hinder academic achievement (i.e., grades)?

Data: “High School And Beyond” survey (1983 wave) ($N = 1375$)

Variables:

- Grades: As=4, As & Bs=3.5, etc.
- Sports: 1 if participated in varsity sports, 0 otherwise
- FamIncome: Family income (7-point scale)
- SES: Socioeconomic Status: 1=low, 2=middle, 3=high
- WorkAge: Age at which started working
- HmwkTime: Time spent on homework (7-point scale)*
- Female: 1 = female student, 0 = male student
- Academic: 1 if the student is on an academic track, 0 else
- Remedial: 1 if the student took ≥ 1 remedial course
- Advanced: 1 if the student took ≥ 1 advanced course

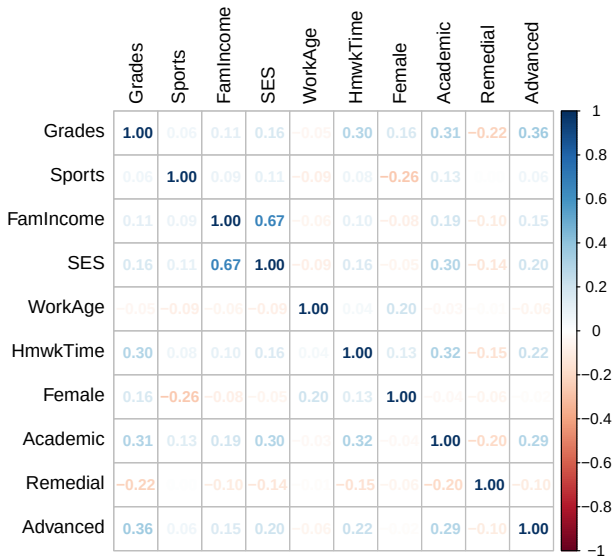
* Likely post-treatment, so we'll omit in the examples below.

Summary Statistics

```
> describe(sports)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Grades	1	1375	2.89	0.77	3	2.94	0.74	0	4	4	-0.60	0.25	0.02
Sports	2	1375	0.37	0.48	0	0.34	0.00	0	1	1	0.53	-1.72	0.01
FamIncome	3	1375	4.37	1.79	5	4.42	1.48	1	7	6	-0.18	-1.01	0.05
SES	4	1375	1.96	0.73	2	1.95	1.48	1	3	2	0.07	-1.13	0.02
WorkAge	5	1375	14.63	2.37	15	14.50	1.48	11	21	10	0.64	0.75	0.06
HmwkTime	6	1375	4.55	1.40	4	4.59	1.48	1	7	6	-0.28	-0.20	0.04
Female	7	1375	0.52	0.50	1	0.53	0.00	0	1	1	-0.08	-1.99	0.01
Academic	8	1375	0.41	0.49	0	0.39	0.00	0	1	1	0.37	-1.86	0.01
Remedial	9	1375	0.36	0.48	0	0.33	0.00	0	1	1	0.58	-1.67	0.01
Advanced	10	1375	0.37	0.48	0	0.34	0.00	0	1	1	0.53	-1.72	0.01

Correlation Plot



Simple *t*-test & OLS Regression

```
> with(sports, t.test(Grades~Sports))

Welch Two Sample t-test

data: Grades by Sports
t = -2.3, df = 1064, p-value = 0.02
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -0.18308 -0.01394
sample estimates:
mean in group 0 mean in group 1
      2.856          2.955

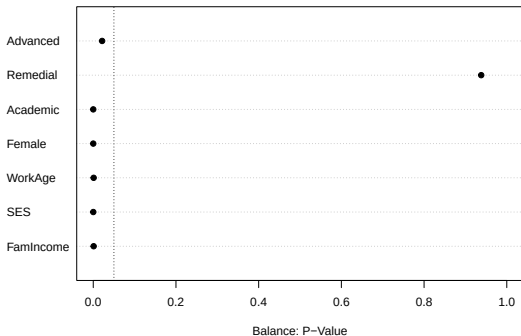
> summary(lm(Model,data=sports))

Residuals:
    Min       1Q   Median       3Q      Max
-2.6938 -0.3761  0.0275  0.4704  1.5020

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.71145    0.13397   20.24 < 2e-16 ***
Sports       0.10119    0.03969    2.55  0.011 *
FamIncome    0.00435    0.01378    0.32  0.753
SES          0.02216    0.03487    0.64  0.525
WorkAge     -0.01879    0.00794   -2.37  0.018 *
Female       0.30062    0.03881    7.75 1.8e-14 ***
Academic     0.29063    0.04099    7.09 2.1e-12 ***
Remedial    -0.23215    0.03919   -5.92 4.0e-09 ***
Advanced     0.44435    0.04004   11.10 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.679 on 1366 degrees of freedom
Multiple R-squared:  0.231, Adjusted R-squared:  0.226
F-statistic: 51.2 on 8 and 1366 DF,  p-value: <2e-16
```

Balance Tests (Pre-Matching)



These are P -values associated with t -tests (for binary predictors) or Kolmogorov-Smirnov tests (for continuous predictors) for balance between `sports = 0` and `sports = 1`.

```
> M.exact <- matchit(Sports~FamIncome+SES+WorkAge+Female+
+                   Academic+Remedial+Advanced,
+                   data=sports,method="exact")
> M.exact
A matchit object
- method: Exact matching
- number of obs.: 1375 (original), 526 (matched)
- target estimand: ATT
- covariates: FamIncome, SES, WorkAge, Female, Academic,
  Remedial, Advanced

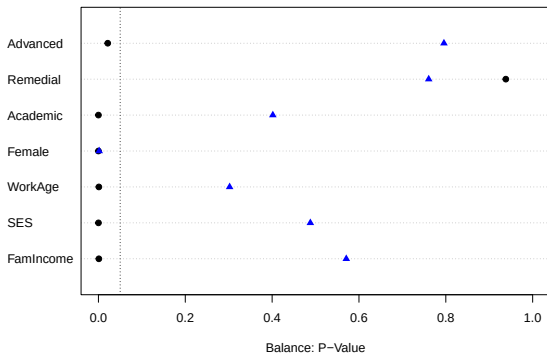
> # Output matched data:

> sports.exact <- match.data(M.exact,group="all")

> dim(sports.exact)

[1] 526  12
```

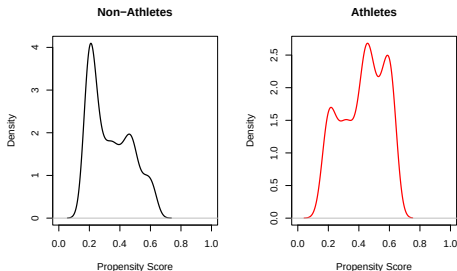
Exact Matching: Balance



These are P -values associated with t -tests (for binary predictors) or Kolmogorov-Smirnov tests (for continuous predictors) for balance between $\text{sports} = 0$ and $\text{sports} = 1$. Black dots are pre-matching; blue triangles are after exact matching.

Propensity Score Matching

```
> PSfit <- glm(Sports~FamIncome+SES+WorkAge+Female+  
+             Academic+Remedial+Advanced,data=sports,  
+             family=binomial(link="logit"))  
  
> # summary(PSfit)  
  
> # Generate scores & check common support:  
  
> PS.df <- data.frame(PS = predict(PSfit,type="response"),  
+                      sports = PSfit$model$Sports)
```



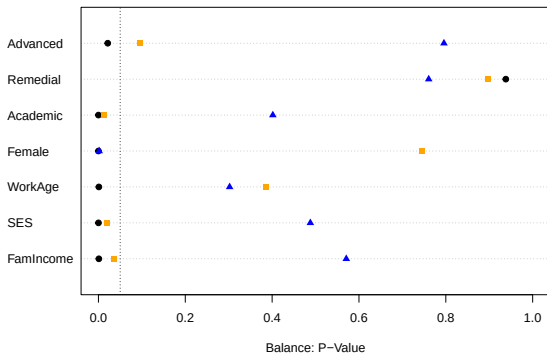
Propensity Score Matching

```
> M.prop <- matchit(Sports~FamIncome+SES+WorkAge+Female+  
+                   Academic+Remedial+Advanced,data=sports,  
+                   method="nearest")  
> summary(M.prop)  
.  
.  
.
```

Sample Sizes:

	Control	Treated
All	864	511
Matched	511	511
Unmatched	353	0
Discarded	0	0

Propensity Score Matching: Balance



These are P -values associated with t -tests (for binary predictors) or Kolmogorov-Smirnov tests (for continuous predictors) for balance between $\text{sports} = 0$ and $\text{sports} = 1$. Black dots are pre-matching; blue triangles are after exact matching; orange squares are after propensity score matching.

Differences in Means

```
> with(sports, t.test(Grades~Sports))$statistic # No matching  
      t  
-2.286
```

```
> with(sports.exact, t.test(Grades~Sports))$statistic # Exact  
      t  
-1.395
```

```
> with(sports.prop, t.test(Grades~Sports))$statistic # PS  
      t  
-3.036
```

```
> with(sports.genetic, t.test(Grades~Sports))$statistic # Genetic  
      t  
-2.542
```

Regression Results

	No Matching	Exact	Propensity Score	Genetic
(Intercept)	2.71* (0.13)	3.05* (0.23)	2.82* (0.16)	2.77* (0.15)
Sports	0.10* (0.04)	0.12* (0.06)	0.10* (0.04)	0.10* (0.04)
Family Income	0.00 (0.01)	0.05 (0.03)	0.00 (0.02)	0.01 (0.02)
SES	0.02 (0.03)	-0.14 (0.07)	0.05 (0.04)	0.03 (0.04)
WorkAge	-0.02* (0.01)	-0.03* (0.01)	-0.03* (0.01)	-0.02* (0.01)
Female	0.30* (0.04)	0.34* (0.06)	0.30* (0.05)	0.31* (0.04)
Academic	0.29* (0.04)	0.24* (0.08)	0.31* (0.05)	0.30* (0.05)
Remedial	-0.23* (0.04)	-0.28* (0.06)	-0.29* (0.05)	-0.26* (0.05)
Advanced	0.44* (0.04)	0.51* (0.08)	0.43* (0.05)	0.42* (0.05)
R ²	0.23	0.29	0.26	0.25
Adj. R ²	0.23	0.28	0.26	0.25
N	1375	526	1022	1022

* $p < 0.05$

Some Questions...

- What – if anything – can the general robustness of our results tell us about the relationship between varsity athletics and grades?
- What can they tell us about our model?
- What mechanism(s) / circumstances might allow us to investigate the relationship between varsity athletic participation and grades using an RDD?
- What circumstances – if any – might allow us to investigate this relationship using instrumental variables?
- What sort(s) of experiments – natural or otherwise – might allow us to investigate this same relationship?

Good references:

- Freedman (2012)*
- Shalizi (someday)*
- Morgan and Winship (2014)
- Pearl et al. (2016)
- Peters et al. (2017)
- Hernan and Robins (2020)*
- Huntington-Klein (2025)
- Yuksel and Aydede (2026)

Courses / syllabi (a sampling):

- Frey (2024)
- Hazlett (2026)*
- Imai (2023)
- Munger (2023)
- Ornstein (2026)*
- Romémont (2026)
- Xu (2018, 2026)
- Yamamoto (2022)

Other useful nuggets:

- The CRAN [task view on causal inference](#)
- Richard McElreath, *Regression, Fire, and Dangerous Things*
- The Credibility Revolution** in Political Science

* I really like this one.

** Used unironically, and without scare quotes!